09-14 第1篇

阅读理解（说明文） Why Do Onions Make Us Cry?

① In many kitchens around the world, the same small drama happens almost every evening. A parent takes out a **fresh white onion**, lifts a kitchen knife, makes the first cut — and within thirty seconds, tears are running down everyone's face nearby. The onion did not shout at us, step on our toe, or say anything rude. Yet we cry. Most people simply accept this small mystery and move on. But behind the tears of a single onion lies a surprisingly clever chemical story worth knowing.

② To understand the tears, we must first understand the **onion itself**. An onion is the underground "food storage" of a small plant. Inside each onion, there are tiny parts called **cells**. Most of the onion is made of two groups: one group stores **sulphur compounds** (sulphur is a chemical element often linked with strong smells), and the other group stores a special **enzyme** (a kind of natural helper that speeds up chemical reactions). On most days, these two groups stay **separated** inside the onion and do not bother anyone.

③ The trouble begins when we **cut** the onion. The knife breaks thousands of cell walls at once. The sulphur compounds and the enzyme, which were quietly stored in different rooms, suddenly mix together. In about **two seconds**, they react and produce a small **gas** — a chemical called, among scientists, **"syn-propanethial-S-oxide"**. It does not stay in the onion. It rises quickly into the air and then into our eyes.

④ When this gas touches the surface of the eye, it reacts slightly with the thin wet layer there and produces a tiny amount of a mild **acid**. The eye immediately recognises this acid as a small danger. A **nerve** near the front of the eye — the same nerve that keeps an eye clean when dust blows in — sends a fast message to the brain: "**Please wash me now.**" The brain answers by ordering the eye's tiny glands to **produce more tears**. These extra tears wash the acid away and protect the eye.

⑤ This is why an onion **does not hurt you**: the tears are a small, well-designed **defence mechanism**, and they do their job well. Within a minute or two, the gas disappears, the eyes dry, and the chopping continues. An onion is, in a sense, never "crying at you". It is your own eye quietly washing itself, following instructions written into your body long before you were born.

⑥ Why does an onion carry such chemicals in the first place? Scientists believe the **gas defence** evolved to protect wild onions from **animals that eat plants**. A rabbit or a hungry insect that bites into a raw onion gets a stinging nose and tears in its eyes, and it

usually chooses another plant next time. In other words, the tears we shed in our kitchen are side effects of a defence meant for a **rabbit 10,000 years ago**. The onion does not know we have a kitchen.

⑦ Now for the good news: there are several simple, science-based ways to **cry less** while cutting onions. **First**, use a **sharp knife**. A sharp knife breaks **fewer cells**, so less gas is released; a blunt knife crushes many more cells and fills the air with a strong cloud. **Second**, **chill the onion in the fridge** for 10—15 minutes before cutting. Cold air holds the gas more lazily, so less of it reaches your eyes. **Third**, cut the onion **next to a running tap or an open window**; moving air carries the gas away. **Fourth**, keep the **root** (the hairy end) attached until last — most of the sulphur compounds are concentrated there.

⑧ Some people enjoy telling stories about onion "magic tricks", such as holding a piece of bread in your teeth or chewing gum while chopping. Scientists have tested several of these tricks; the results are **mixed**. They may help a little by changing the way you breathe, but they are not as reliable as a sharp knife and a cold onion. Fancy "onion goggles" do work, if you think they look cool enough to wear in your own kitchen.

⑨ So next time a thin, sharp smell rises from a cutting board and your eyes fill with small unexpected tears, you know the full story. No onion is being cruel. Two chemical groups, separated for weeks, have finally met. One small gas has travelled a few centimetres to the surface of your eye. One careful body has reacted exactly as it should. All these quiet events, repeated in millions of kitchens each night, add up to one small, warm truth: **our eyes have been taking care of us the whole time, even over a plate of noodles**.

注释： sulphur 硫；enzyme 酶；acid 酸；nerve 神经；blunt knife 钝刀

根据文章内容，选择最佳答案。

1. Inside an onion, what two things are stored separately until the onion is cut?

A. Water and sugar.
B. Sulphur compounds and a special enzyme.
C. Salt and oil.
D. Sand and soap.

2. According to paragraph 3, what happens about two seconds after you cut an onion?

A. The onion becomes completely dry.
B. The two groups of chemicals mix and produce a gas that rises into the air.
C. The onion turns red immediately.
D. The enzyme flies out of the kitchen window.

3. Why does the eye react by producing extra tears?

A. Because the gas from the onion is beautiful.
B. Because a small amount of mild acid forms on the eye, and a nerve sends a message to the brain to "wash the eye".
C. Because the knife is usually dirty.
D. Because crying makes people dance.

4. Why did the "tear gas" defence of onions first evolve, according to the passage?

A. To entertain human chefs.
B. To protect wild onions from animals that eat plants, such as rabbits and insects.
C. To scare away butterflies.
D. To help onions grow taller.

5. Which of the following is NOT one of the practical tips for crying less while cutting onions?

A. Use a sharp knife to break fewer cells.
B. Chill the onion in the fridge for 10—15 minutes first.
C. Cut next to a running tap or an open window.
D. Shout angrily at the onion before starting.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "one group stores sulphur compounds ... the other group stores a special enzyme"
【解析】硫化合物与酶被隔开储存。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "In about two seconds, they react and produce a small gas"
【解析】切开后两秒左右，两者反应产生气体升起。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "a mild acid ... A nerve ... sends a fast message to the brain: 'Please wash me now.'"
【解析】酸刺激引起神经反射性地增多泪液以清洗眼睛。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "the gas defence evolved to protect wild onions from animals that eat plants"
【解析】原为防食草动物而演化。A、C、D 均错。故选 B。

5.
【答案】D
【题型】细节判断题（否定型）
【定位】第⑦段四条建议
【解析】文章未建议"对洋葱大喊"。A、B、C 均为其建议。故选 D。